Building a Bridge of Bounding Box Regression Between Oriented and Horizontal Object Detection in Remote Sensing Images

Xiaoliang Qian, *Member, IEEE*, Baokun Wu, Gong Cheng, *Member, IEEE*, Xiwen Yao, *Member, IEEE*, Wei Wang, and Junwei Han, *Fellow, IEEE*

***Abstract*— Oriented object detection (OOD) aims to precisely detect the objects with arbitrary orientation in remote sensing images (RSIs). Up to now, most of the bounding box regression (BBR) losses for OOD are transferred from horizontal object detection (HOD) methods; however, the transferring requires lots of professional knowledge and experiences for designers, and consequently, many excellent BBR losses for HOD have not been transferred to OOD. To accelerate the research progress of BBR loss for OOD, a unified transferring strategy (UTS) is proposed to facilitate the transferring of BBR loss from HOD to OOD. The UTS proposes that the BBR of oriented bounding box (OBB) can be converted into the joint BBR of its horizontal smallest enclosing rectangle (HSER) and two offsets, so the BBR loss in HOD can be easily transferred to OOD using HSER as a bridge (BR). Following the UTS, a BBR loss named rotated-intersection of union (RIoU) loss is designed for OOD by transferring an advanced BBR loss in HOD, which can be considered as an example to show how to transfer. On the basis of RIoU loss, a focal RIoU (FRIoU) loss is proposed to assign larger weights to hard samples in the BBR. The comparisons with other BBR losses show that the RIoU and FRIoU losses can give better performance. The ablation study shows that giving more attention to hard samples in BBR is effective. The comparisons with many advanced methods demonstrate that the combinations of baseline methods and FRIoU loss achieve state-of-the-art (SOTA) performance on the DOTA and DIOR-R datasets.**

***Index Terms*— Bounding box regression (BBR), focal rotated intersection of union (FRIoU) loss, oriented object detection (OOD), remote sensing images (RSIs), unified transferring strategy (UTS).**

I. Introduction

COMPARED with horizontal object detection (HOD), oriented object detection (OOD) can precisely detect the objects (such as bridges (BRs), ports, oil storage tanks) with arbitrary orientation in remote sensing images (RSIs) [1], [2], [3], [4], [5], which plays a crucial role in RSIs' interpretation and can be applied in intelligent monitoring, urban planning, precision agriculture, geographic information system updating, etc.

Up to now, deep-learning-based object detection methods [6], [7], [8], [9], [10], [11], [12], [13], [14], [15] are mainstream, the training losses of which generally consist of classification and bounding box regression (BBR) losses. The classification losses are usually designed based on cross-entropy, such as focal loss [16], GFLV1 [17], GFLV2 [18], and varifocal loss [19], and the existing designing ideas have no major changes. The study of BBR loss is still a hot topic, and many innovative works [20], [21] are proposed. The existing BBR losses for HOD and OOD are briefly summarized as follows, respectively.

The Smooth $L1$ loss proposed by Fast R-CNN [22] is a milestone work and adopted by many HOD methods for BBR, which combines the advantages of $L1$ loss and $L2$ loss. Specifically, the $L1$ loss is used for linear optimization when the difference between the predicted box and the ground-truth box is large, and the $L2$ loss is adopted to gradually reduce the learning speed when the difference is small. The implicit premise of Smooth $L1$ loss is that the four spatial coordinates are independent of each other; however, the premise is invalid in practical applications. To address this problem, the intersection of union (IoU) loss [23] was proposed to directly optimize the IoU between the predicted box and the ground-truth box; however, it is invalid when the predicted box and the ground-truth box are nonintersecting. Afterward, the GIoU loss [24] was proposed to address the problem by introducing the smallest enclosing rectangle (SER) of the two bounding boxes. Subsequently, the CIoU loss [25] pointed out that the GIoU loss was invalid when one box was completely enclosed by another bounding box, and thus, the distance between the centers of the predicted and ground-truth boxes, and the aspect ratio of height and width were incorporated into the GIoU loss. Recently, the EIoU loss [26] indicated that the aspect ratio was not rigorous and replaced it by directly optimizing the deviations of height and width. Other important BBR losses include Alpha-IoU [20], DIoU [25], etc.

At present, most of the BBR losses of OOD are transferred from HOD methods [27], [28], [29]. The Smooth $L1$ loss was adopted by many OOD methods [30], [31], [32], similarly, lots of IoU-based BBR losses were also used to replace the Smooth

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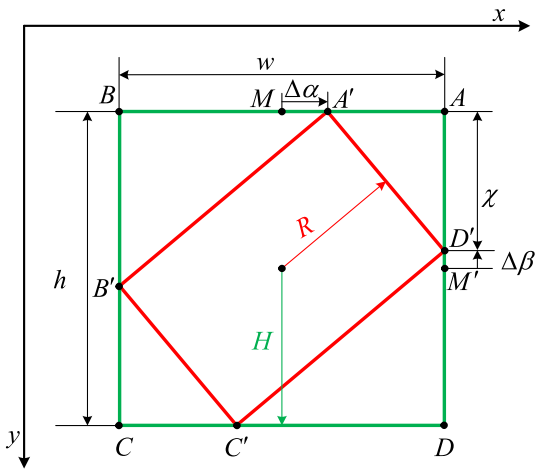


Fig. 1. Illustration of horizontal representation of OBB. R ($\square A'B'C'D'$) denotes OBB, H ($\square ABCD$) denotes HSER of R, M (M') denotes the midpoint of AB (AD), $\Delta\alpha$ ($\Delta\beta$) denotes the offset of point A' (D') relative to M (M'), χ denotes the length of $A'D'$, and w and h denote the width and height of H, respectively. R can be solely determined by H, $\Delta\alpha$, and $\Delta\beta$.

$L1$ loss. The IoU loss, GIoU loss, and CIoU loss used for HOD were improved to adjust to OOD [33], [34]. Chen et al. [35] proposed PIoU loss on the basis of IoU loss, where the number of pixels in the intersection region of two bounding boxes was calculated through a kernel-function-based model and was considered as the area of intersection region. Yang et al. [28] and [29] also improved the computational scheme of IoU loss, which measured the overlapping degree of two bounding boxes through the Gaussian distance. Other important BBR losses of OOD include projection IoU [36], SCRDet [37], etc.

For now, the transferring of BBR loss from HOD to OOD requires lots of professional knowledge and experiences for designers, and consequently, many excellent BBR losses for HOD have not been transferred to OOD. To accelerate the research progress of BBR loss for OOD, a unified transferring strategy (UTS) of BBR loss from HOD to OOD is proposed in this article, and it is used to design a novel BBR loss named focal rotated IoU (FRIoU) loss. First, as shown in Fig. 1, inspired by Oriented R-CNN [30], the UTS points out that the oriented bounding box (OBB) R can be solely determined by its horizontal SER (HSER) H and two offsets ($\Delta\alpha$ and $\Delta\beta$) (related proof can be seen in Section III-A2), and consequently, the BBR of R can be converted into the joint BBR of H, $\Delta\alpha$, and $\Delta\beta$, and thus, the excellent BBR losses in HOD can be converted into OOD with H as a BR. Afterward, according to the aforementioned UTS, a BBR loss named rotated-IoU (RIoU) loss is designed by transferring the EIoU loss [26] in HOD, it and is also used as an example to show the processing of transferring. Finally, on the basis of RIoU loss, the FRIoU loss is proposed to assign larger weights to hard samples in the processing of BBR.

The major contributions of this article can be summarized as follows.

1) A UTS is proposed to provide a universal solution which facilitates the transferring of BBR loss from HOD to OOD.
2) Following the proposed UTS, an IoU-based BBR loss named RIoU loss is designed for OOD by transferring the EIoU loss in HOD. It is worth noting that other BBR losses for HOD can also be transferred to OOD following the UTS, and RIoU loss can be considered as an example to show how to transfer.
3) On the basis of RIoU loss, a novel BBR loss named FRIoU loss is proposed to give more attention to hard samples in the processing of BBR.
4) The comparisons with other BBR losses and comprehensive comparisons on the DOTA and DIOR-R datasets validate the effectiveness of the proposed UTS and FRIoU loss, and the good generality of FRIoU loss.

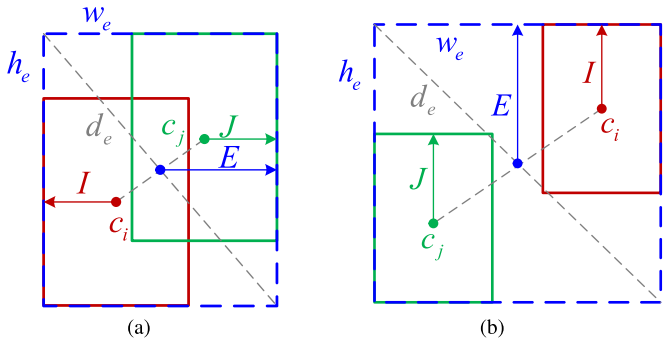


Fig. 2. Illustration of two (a) intersecting and (b) nonoverlapping bounding boxes and their SER.

II. Selecting IoU-Based BBR Loss in HOD

A suitable BBR loss in HOD which is used for transferring to OOD is selected in this section. Although the Smooth $L1$ loss is adopted by many HOD methods, the implicit premise of Smooth $L1$ loss is that the four spatial coordinates are independent of each other; obviously, the premise is invalid in practical applications. Therefore, a brief analysis of IoU-based BBR losses in HOD is presented to select the appropriate one used for transferring to OOD.

The IoU is used to evaluate the overlapping degree of any two horizontal bounding boxes (HBBs) I and J

$$\text{IoU} = \frac{|I \bigcap J|}{|I \bigcup J|} \tag{1}$$

where $|\cdot|$ denotes the cardinality of assemble.

The formulation of IoU loss (L_{IoU}) [23] is as follows:

$$L_{\text{IoU}} = 1 - \text{IoU}. \tag{2}$$

As shown in Fig. 2(b), the IoU is always zero when I and J are nonoverlapping; consequently, the IoU loss cannot optimize the model parameters at this time. The GIoU loss (L_{GIoU}) [24] addressed this problem by introducing the SER of I and J, denoted as E

$$L_{\text{GIoU}} = 1 - \text{IoU} + \frac{|E - (I \bigcup J)|}{|E|}. \tag{3}$$

However, the GIoU loss also cannot optimize the model parameters when I or J is completely enclosed by another one. The DIoU loss (L_{DIoU}) [25] addressed the above problem by introducing the distance between c_i and c_j

$$L_{\text{DIoU}} = 1 - \text{IoU} + \frac{\rho^2(c_i, c_j)}{d_e^2} \tag{4}$$

where c_i and c_j denote the center points of I and J, respectively, $\rho(\cdot,\cdot)$ denotes the Euclidean distance of two points, and d_e denotes the length of the diagonal of E.

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The CIoU loss (L_{CIoU}) [25] further enhanced the DIoU loss by adding a constraint term about the aspect ratio of bounding boxes

$$\begin{aligned} L_{\mathrm{CIoU}} &= 1 - \mathrm{IoU} + \frac{\rho^2(c_i, c_j)}{d_e^2} + \mu\nu \\ \nu &= \frac{4}{\pi^2}\left(\arctan\frac{w_i}{h_i} - \arctan\frac{w_j}{h_j}\right)^2 \\ \mu &= \frac{\nu}{(1-\mathrm{IoU}) + \nu} \end{aligned} \tag{5}$$

where w_i and h_i (w_j and h_j) denote the width and height of I (J), respectively, ν denotes the deviation of the aspect ratio of I and J, and μ denotes the weight of ν.

However, Zhang et al. [26] proposed that the same aspect ratio was not equivalent to the same width and height, and it was sometimes an obstacle for BBR instead, and therefore, they proposed the EIoU Loss (L_{EIoU}) to handle this problem, which replaced the deviation of the aspect ratio with the deviation of the width and height of I and J

$$L_{\mathrm{EIoU}} = 1 - \mathrm{IoU} + \frac{\rho^2(c_i, c_j)}{d_e^2} + \frac{(w_i - w_j)^2}{w_e^2} + \frac{(h_i - h_j)^2}{h_e^2} \tag{6}$$

where w_e and h_e denote the width and height of E, respectively.

Up to date, the designing scheme of EIoU loss is more comprehensive, and its performance is excellent on HOD benchmarks, and therefore, the EIoU loss is transferred to the OOD in this article. Of course, other excellent IoU based-BBR losses in HOD can also be used for transferring.

III. Proposed Method

The UTS of BBR loss from HOD to OOD is first introduced in this section. Afterward, the EIoU loss is transformed into RIoU loss according to the proposed UTS. Finally, the FRIoU loss is proposed on the basis of RIoU loss to focus on hard samples in BBR.

A. UTS of BBR Loss

1) Unified Transferring Strategy: At present, many BBR losses in HOD are only suitable for HBB, and it is difficult for them to be directly transferred to the BBR of OBB, especially for constructing the SER of two OBBs, such as GIoU, DIOU, CIoU, and EIoU losses. To address this situation, the UTS is proposed to convert the BBR of OBB into the BBR of HBB. As shown in Fig. 1, inspired by Oriented R-CNN [30], the UTS points out that OBB R can be solely determined by its HSER H and two offsets ($\Delta\alpha$ and $\Delta\beta$), so the core idea of UTS is that the BBR of R can be converted into the joint BBR of H, $\Delta\alpha$, and $\Delta\beta$, and thus, H can be considered as a BR between OOD and HOD for BBR; consequently, all the existing BBR losses in HOD can be easily transferred to OOD through H. Obviously, the precondition of using UTS is the proof of one-to-one correspondence between R and H with two offsets. The details of proof are as follows.

2) Proof of One-to-One Correspondence Between OBB and Its Horizontal Minimum Enclosing Rectangle With Two Offsets: As shown in Fig. 1, it is easy to prove $\Delta A'BB' \sim \Delta D'AA'$, which can conclude that

$$\frac{A'B}{D'A} = \frac{BB'}{AA'}. \tag{7}$$

Intuitively, OBB R can be solely determined by its HSER and one offset. Assume that $\Delta\alpha$ is known and $\Delta\beta$ is unknown, consequently, $A'B = w/2 + \Delta\alpha$ ($\Delta\alpha > 0$ in Fig. 1), $A'A = w/2 - \Delta\alpha$ and $BB' = DD' = h - \chi$ because $\triangle A'BB' \cong \triangle C'DD'$, then (7) can be changed to

$$\frac{(w/2 + \Delta\alpha)}{\chi} = \frac{h - \chi}{(w/2 - \Delta\alpha)} \tag{8}$$

$$\Rightarrow \chi^2 - h\chi + (w/2)^2 - \Delta\alpha^2 = 0 \tag{9}$$

$$\chi_1 = \frac{h - \sqrt{h^2 - w^2 + 4\Delta\alpha^2}}{2}$$

$$\chi_2 = \frac{h + \sqrt{h^2 - w^2 + 4\Delta\alpha^2}}{2}. \tag{10}$$

It is easy to conclude that both χ_1 and χ_2 are valid solutions of χ under the condition of $|\Delta\alpha| \le w/2$; in other words, R cannot be solely determined by H and $\Delta\alpha$.

Assume that both $\Delta\alpha$ and $\Delta\beta$ are known, it is easy to conclude that $B'C = D'A = w/2 + \Delta\beta$ ($\Delta\beta < 0$ in Fig.1), and $CC' = AA' = w/2 - \Delta\alpha$ because $\triangle D'AA' \cong \triangle B'CC'$. In summary, the positions of A', B', C', and D' on AB, BC, CD, and DA can be solely determined by w, h, $\Delta\alpha$, and $\Delta\beta$, i.e., R can be solely determined by H, $\Delta\alpha$, and $\Delta\beta$.

B. RIoU Loss

As mentioned in Section II, the IoU-based BBR losses in HOD are briefly analyzed, and the suitable one (EIoU loss) among them is selected to be transferred to OOD. Therefore, the transferred BBR loss is named as RIoU loss. Note that EIoU loss is not the only choice for transferring, other IoU-based BBR losses are also practicable. As mentioned in Section III-A2, the BBR of any OBB is equivalent to the joint BBR of its HSER and two offsets, and therefore, the RIoU loss, denoted as L_{RIoU}, is defined as

$$L_{\mathrm{RIoU}} = 1 - \mathrm{RIoU} = L_H + L_D \tag{11}$$

where L_H and L_D denote the BBR loss of the HSER and offsets, respectively, and the RIoU can be considered as a novel metric which is similar to IoU, GIoU, etc.

As shown in Fig. 3, the formulation of L_D is defined as

$$\begin{aligned} L_D &= \left(\frac{\Delta\alpha_p}{w_p} - \frac{\Delta\alpha_g}{w_g}\right)^2 + \left(\frac{\Delta\beta_p}{h_p} - \frac{\Delta\beta_g}{h_g}\right)^2 \\ \Delta\alpha_p &= x_{n_p} - x_{m_p},\ \Delta\alpha_g = x_{n_g} - x_{m_g} \\ \Delta\beta_p &= y_{n'_p} - y_{m'_p},\ \Delta\beta_g = y_{n'_g} - y_{m'_g} \end{aligned} \tag{12}$$

where m_p (m'_p) and m_g (m'_g) denote the midpoints of the upper borders (right borders) of P and G, respectively, n_p (n'_p) denotes the intersection of P_o and the upper borders (right borders) of P, and n_g (n'_g) denotes the intersection of G_o and the upper borders (right borders) of G, x_{m_p} ($x_{m'_p}$), x_{n_p} ($x_{n'_p}$), x_{m_g} ($x_{m'_g}$), and x_{n_g} ($x_{n'_g}$) denote the coordinates of m_p (m'_p), n_p (n'_p), m_g (m'_g), and n_g (n'_g) in the x-axis (y-axis),

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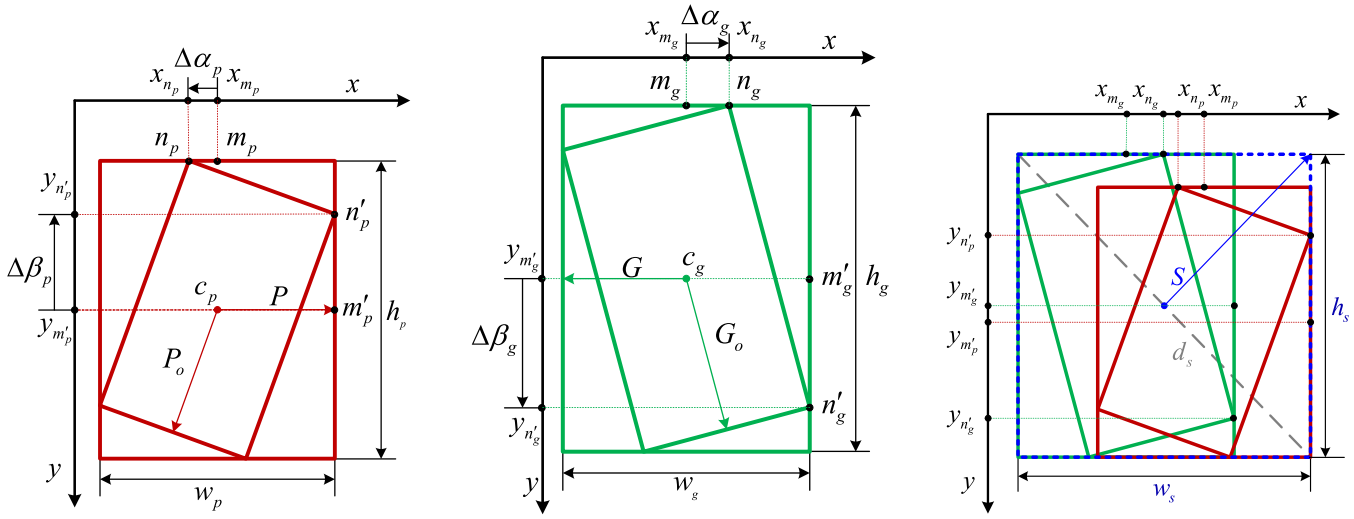


Fig. 3. Graphic indication of RIoU loss. (a) P_o (predicted OBB) and P (HSER of P_o), (b) G_o (ground-truth OBB) and G (HSER of G_o), and (c) S (HSER of P and G).

respectively, and $\Delta\alpha_p$ ($\Delta\alpha_g$) denotes the offset of n'_p (n'_g) relative to m'_p (m'_g).

The formulation of L_H is inherited from the EIoU loss [26]

$$L_H = 1 - \text{IoU} + \frac{\rho^2(c_p, c_g)}{d_s^2} + \frac{(w_p - w_g)^2}{w_s^2} + \frac{(h_p - h_g)^2}{h_s^2} \tag{13}$$

where c_p and c_g denote the centers of P and G, respectively. The (w_p, h_p), (w_g, h_g), and (w_s, h_s) denote the width and height of P, G, and S, respectively.

The final formulation of L_{RIoU} can be obtained by substituting (12) and (13) into (11)

$$L_{\text{RIoU}} = 1 - \text{IoU} + \frac{\rho^2(c_p, c_g)}{d_s^2} + \frac{(w_p - w_g)^2}{w_s^2} + \frac{(h_p - h_g)^2}{h_s^2} + \left(\frac{\Delta\alpha_p}{w_p} - \frac{\Delta\alpha_g}{w_g}\right)^2 + \left(\frac{\Delta\beta_p}{h_p} - \frac{\Delta\beta_g}{h_g}\right)^2 \tag{14}$$

$$\text{RIoU} = \text{IoU} - \frac{\rho^2(c_p, c_g)}{d_s^2} - \frac{(w_p - w_g)^2}{w_s^2} - \frac{(h_p - h_g)^2}{h_s^2} - \left(\frac{\Delta\alpha_p}{w_p} - \frac{\Delta\alpha_g}{w_g}\right)^2 - \left(\frac{\Delta\beta_p}{h_p} - \frac{\Delta\beta_g}{h_g}\right)^2. \tag{15}$$

C. FRIoU Loss

As shown in (11), the RIoU loss is a linear function of RIoU, so its gradient value always remains constant, which indicates that all the samples are given the same weight in the processing of BBR. However, as mentioned in focal loss [16], giving more attention to hard samples can improve the performance of object detection model, and therefore, the absolute value of gradient of the BBR loss should be inversely proportional to the value of RIoU. Following the above analysis, the FRIoU loss, denoted as L_{FRIoU}, is proposed on the basis of RIoU loss:

$$L_{\text{FRIoU}} = e^{(-\lambda \text{RIoU})} L_{\text{RIoU}} \tag{16}$$

where λ is a hyperparameter that controls the change rate of gradient. As shown in Fig. 4, the absolute value of gradient of the FRIoU loss is gradually increased with the decrease in RIoU; in other words, the hard samples are given more attention.

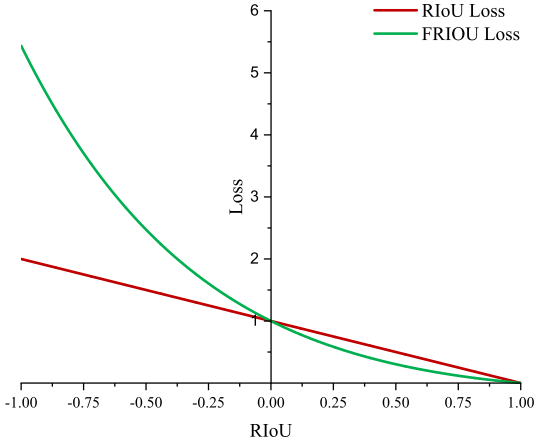


Fig. 4. Illustration of RIoU and FRIoU losses.

IV. Experiments

A. Experiment Setup

1) Datasets: DOTA [38][1] is a large-scale aerial RSI dataset which contains 2806 images and 188 282 instances with 15 common object categories: plane (PL), baseball diamond (BD), BR, ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ballfield (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). For single-scale training and testing, the original image is cropped into a number of subimages with the size of 1024 × 1024 pixels by a step of 824 pixels. For multiscale training and testing, the original image is first resized with three scales (0.5, 1.0, 1.5), and then is cropped into a number of subimages with the size of 1024 × 1024 pixels by a step of 524 pixels. All the testing results are obtained from the DOTA evaluation server.[2]

[1]https://captain-whu.github.io/DOTA/index.html
[2]http://bed4rs.net:8001/evaluation1

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TABLE I
COMPARISONS WITH OTHER BBR LOSSES AND ABLATION STUDY ON THE DOTA AND DIOR-R DATASETS. THE BOLD FONTS DENOTE THE BEST RESULTS

Method	Backbone	Epoch	BBR Loss	mAP	
				DOTA	DIOR-R
AOPG	R-50-FPN	12	Smooth L1 Loss	75.24	64.41
			IoU Loss	75.51	65.29
			GIoU Loss	75.56	65.80
			PIoU Loss	73.73	63.35
			RIoU Loss	76.00	66.51
			FRIoU Loss	**76.45**	**67.42**

The DIOR-R dataset [39][3] was recently proposed for OOD, which shared the same images with previous DIOR dataset [40] and relabeled each instance with an OBB. It includes 23 463 images and 192 518 instances with 20 common object categories: airplane (APL), airport (APO), baseball field (BF), BC, BR, chimney (CH), expressway service area (ESA), expressway toll station (ETS), dam (DAM), golf field (GF), GTF, HA, overpass (OP), SH, stadium (STA), storage tank (STO), TC, train station (TS), vehicle (VE), and windmill (WM). The size of each image is 800 × 800 pixels.

Both the training and validation sets of the above two datasets are used for training, and the rest is used for testing. For fair comparison and preventing overfitting, the randomly horizontal and vertical flipping is adopted on both the datasets during training [2], [39].

2) Implementation Details:

1) *Training Details:* The entire OOD model is optimized by the stochastic gradient descent (SGD) algorithm with a momentum of 0.9 and a weight decay of 0.0001. The batchsize and initial learning rate are the same as the baseline methods. For 12 (36) epochs' training, the learning rate is decayed to 1/10 of the last stage at the eighth (24th) and 11th (33th) epochs. The threshold of non-maximum suppression (NMS) is usually set to 0.1 [38] and 0.5 [39] on the DOTA and DOIR-R datasets, respectively.
2) *Environment Settings:* The experiments are conducted on mmdetection [41][4] within the PyTorch framework and run on a workstation with two Intel[5] Xeon[5] CPU E5-2630 v4 (a total of 10 × 2 cores with 2.20 GHz), 256-GB memory, and eight NVIDIA GeForce RTX 2080 Ti (a total of 11 × 8-GB memory).
3) *Evaluation Metrics:* The average precision (AP) of each category and mean average precision (mAP) of all the categories are adopted as the metrics to evaluate the performance of the proposed method.

B. Parameter Analysis

The λ in (16) is the only hyperparameter in this article, and it is quantitatively analyzed in this section to evaluate its impact. As shown in Fig. 5, the best results are all obtained at $\lambda = 1$ on the DOTA and DIOR-R datasets, which demonstrates the robustness of the proposed method.

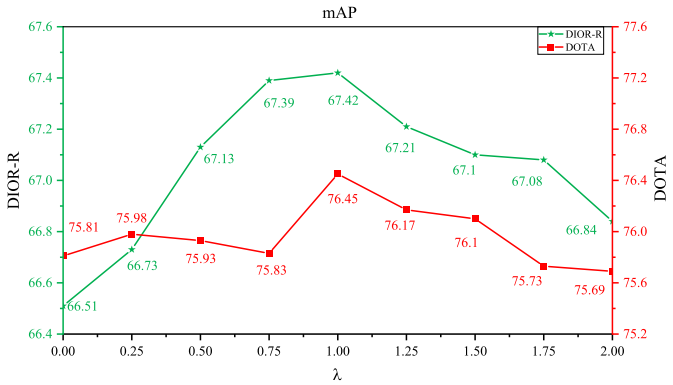


Fig. 5. Quantitative parameter analysis of λ on the DOTA and DIOR-R datasets. The AOPG [39] is used as the baseline method, of which the backbone network is R-50-FPN [42].

C. Comparisons With Other BBR Losses and Ablation Study

To evaluate the effectiveness of the proposed UTS, the RIoU and FRIoU losses are compared with other BBR losses adopted by the existing OOD methods, which include Smooth $L1$ loss [43], [44], [45], PIoU loss [35], IoU loss, and GIoU loss [33]. In addition, the comparisons between RIoU and FRIoU loss are considered as the ablation study of FRIoU loss. The AOPG [39] is adopted as the baseline method, of which the backbone network is R-50-FPN (denotes ResNet-50-FPN [42]), and the number of training epochs is set to 12.

As shown in Table I, the experimental results of the proposed RIoU and FRIoU losses are superior to other BBR losses, which demonstrates the effectiveness of the proposed UTS. Furthermore, the ablation study of FRIoU loss indicates that giving more attention to hard samples in the BBR can effectively improve the performance of OOD methods.

D. Validation of Generalization Capability

To validate the generalization capability of FRIoU loss, it is used to replace the original BBR losses of six advanced OOD methods, i.e., AOPG [39], S^2A-Net [2], RoI Trans. [31], Oriented R-CNN [30], Faster-RCNN-O [46], and RetinaNet-O [16], and then the six OOD methods are compared with their replacing versions. The six OOD methods include anchor-free and anchor-based methods, and one-stage and two-stage methods. The backbone of six OOD methods is R-50-FPN, and the number of training epochs is set to 12.

As shown in Table II, the mAP of AOPG, S^2A-Net, RoI Trans., Oriented R-CNN, Faster-RCNN-O, and RetinaNet-O separately increased by 1.21%, 1.06%, 1.71%, 0.50%, 3.49%, and 1.03% on the DOTA dataset via using FRIoU loss and separately increased by 3.01%, 0.48%, 0.91%, 0.42%, 1.69%, and 1.19% on the DIOR-R dataset. The above experimental

[3]https://gcheng-nwpu.github.io/DIOR-R
[4]https://github.com/open-mmlab/mmdetection
[5]Registered trademark.

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TABLE II

COMPARISONS BETWEEN SIX ADVANCED OOD METHODS AND THEIR REVISED VERSIONS IN WHICH THE ORIGINAL BBR LOSSES ARE REPLACED WITH FRIOU LOSS. * DENOTES THE RESULTS FROM AERIAL DETECTION. THE SAME BELOW

Method	Backbone	Epoch	FRIoU	mAP	
				DOTA	DIOR-R
AOPG	R-50-FPN	12		75.24	64.41
			√	76.45 (+1.21)	67.42 (+3.01)
S^2A-Net				74.12	62.90
			√	75.18 (+1.06)	63.38 (+0.48)
RoI Trans.*				74.61	63.87
			√	76.32 (+1.71)	64.78 (+0.91)
Oriented R-CNN				75.87	64.63
			√	76.37 (+0.50)	65.15 (+0.62)
Faster-RCNN-O*				69.05	59.04
			√	72.54 (+3.49)	61.23 (+1.69)
RetinaNet-O*				68.43	57.55
			√	69.46 (+1.03)	58.74 (+1.19)

TABLE III

COMPARISONS WITH ADVANCED OOD METHODS ON THE DOTA DATASET. THE PROPOSED FRIOU LOSS IS EMBEDDED IN S^2A-NET [2] AND AOPG [39] FOR COMPARISONS. THE DETECTION RESULTS OF EACH CATEGORY ARE REPORTED TO BETTER DEMONSTRATE WHERE THE PERFORMANCE GAINS COME FROM. THE BLUE AND RED FONTS DENOTE THE BEST RESULTS AMONG THE ONE-STAGE AND TWO-STAGE METHODS, RESPECTIVELY. ‡ DENOTES MULTISCALE TRAINING AND TESTING

Methods	Backbone	Epoch	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
One-stage:																		
PIoU [35]	DLA-34	12	80.90	69.70	24.10	60.20	38.30	64.40	64.80	90.90	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
RetinaNet-O* [16]	R-50-FPN	12	88.67	77.62	41.81	58.17	74.58	71.64	79.11	90.29	82.18	74.32	54.75	60.60	62.57	69.57	60.64	68.43
DRN [47]	H-104	120	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.6	52.50	70.70
LO-Det-GGHL [48]	MobileNetv2	36	89.66	83.02	38.55	77.09	72.57	71.86	82.47	90.78	78.05	83.56	47.74	67.83	64.21	67.83	54.16	71.26
S^2A-Net [2]	R-50-FPN	12	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
Oriented RepPoints [43]	R-50-FPN	40	87.02	83.17	54.13	71.16	**80.18**	78.40	87.28	**90.90**	85.97	86.25	59.90	70.49	73.53	72.27	58.97	75.97
Ours:																		
S^2A-Net + FRIoU	R-50-FPN	12	88.77	80.79	51.75	74.48	78.86	78.92	87.56	90.88	86.01	84.78	61.11	66.61	68.61	69.24	59.32	75.18
S^2A-Net + FRIoU	R-50-FPN	36	89.01	83.23	53.54	73.14	79.09	81.23	87.94	90.88	**86.68**	83.63	62.55	67.09	74.87	70.65	62.62	76.38
S^2A-Net + FRIoU‡	R-50-FPN	12	**90.09**	84.04	**58.45**	81.97	79.25	84.00	88.50	90.84	84.98	87.21	**72.50**	71.66	79.32	81.20	**71.87**	80.39
S^2A-Net + FRIoU‡	R-50-FPN	36	89.83	**85.08**	56.17	**83.11**	79.95	**84.41**	**88.78**	90.78	85.66	**87.70**	72.05	**73.08**	**81.54**	**81.64**	71.23	**80.74**
Two-stage:																		
Faster-RCNN-O* [46]	R-50-FPN	12	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05
CenterMap-Net [49]	R-50-FPN	12	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.70	71.74
SCRDet [37]	R-101-FPN	12	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
RoI Trans.* [31]	R-50-FPN	12	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
DODet [50]	R-50-FPN	12	89.34	84.31	51.39	71.04	79.04	82.86	88.15	90.90	86.88	84.91	62.69	67.63	75.47	72.22	45.54	75.49
Oriented R-CNN [30]	R-50-FPN	12	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
AOPG [39]	R-50-FPN	12	89.27	83.49	52.50	69.97	73.51	82.31	87.95	90.89	87.64	84.71	60.01	66.12	74.19	68.30	57.80	75.24
AOPG‡ [39]	R-50-FPN	12	89.88	85.57	**60.90**	**81.51**	78.70	85.29	**88.85**	90.89	87.60	87.65	**71.66**	68.69	82.31	77.32	73.10	80.66
Ours:																		
AOPG + FRIoU	R-50-FPN	12	89.94	82.26	54.55	70.69	78.71	83.38	88.00	90.83	86.33	84.38	65.66	66.42	74.87	69.05	61.64	76.45
AOPG + FRIoU	R-50-FPN	36	89.09	84.64	54.46	76.07	78.87	83.99	88.08	90.89	85.43	83.41	66.12	68.91	76.87	72.97	54.68	76.96
AOPG + FRIoU‡	R-50-FPN	12	89.95	84.86	59.62	81.34	78.63	85.24	88.46	**90.90**	**88.10**	**87.66**	69.85	71.47	82.29	77.94	**75.41**	80.78
AOPG + FRIoU‡	R-50-FPN	36	**90.13**	**85.85**	60.04	80.90	**79.21**	**85.34**	88.73	90.88	86.35	87.55	70.22	**72.09**	**83.40**	**82.12**	72.55	**81.02**

results demonstrate that the FRIoU loss has a good generalization capability for various types of OOD methods.

E. Comprehensive Comparisons on the DOTA and DIOR-R Datasets

To fully validate the effectiveness of FRIoU loss, S^2A-Net [2] and AOPG [39] are selected as the one-stage and two-stage baseline methods, respectively, and the FRIoU loss is embedded in them, denoted as S^2A-Net + FRIoU and AOPG + FRIoU, respectively. As shown in Tables III and IV, they are quantitatively compared with other advanced one-stage and two-stage OOD methods on the DOTA and DIOR-R datasets, respectively. The related experimental results and analysis are as follows.

1) Experiment Results on the DOTA Dataset: For the comparisons of the one-stage methods, as shown in Table III, the mAP of S^2A-Net + FRIoU (epoch = 36) is higher than Oriented RepPoints [43] (epoch = 40) which is the state-of-the-art (SOTA) one-stage method, and the mAP of S^2A-Net + FRIoU (epoch = 12) is higher than S^2A-Net [2] (epoch = 12) which is the SOTA one-stage method when epoch = 12. For the comparisons of the two-stage methods, the mAP of AOPG + FRIoU (epoch = 12) is higher than Oriented R-CNN [30] (epoch = 12) which is the SOTA two-stage method. For multiscale training and testing, the mAP of AOPG + FRIoU (epoch = 12) is higher than AOPG (epoch = 12), and the mAP of S^2A-Net + FRIoU (epoch = 36) and AOPG + FRIoU (epoch = 36) achieves pretty high scores, i.e., 80.74% and 81.02%. Furthermore, S^2A-Net + FRIoU obtains the best results in 13 out of 15 categories, and AOPG + FRIoU obtains the best results in 11 out of 15 categories.

2) Experiment Results on the DIOR-R Dataset: For the comparisons of the one-stage methods, as shown in Table IV, the mAP of S^2A-Net + FRIoU (epoch = 36) is higher than Oriented RepPoints [43] (epoch = 40) which is the SOTA one-stage method, and the mAP of S^2A-Net + FRIoU (epoch = 12) is higher than S^2A-Net [2] (epoch = 12) which is the SOTA one-stage method when epoch = 12. For the comparisons of the two-stage methods, the mAP of AOPG + FRIoU (epoch = 12) is higher than DoDet [50] (epoch = 12) which is the SOTA two-stage method. Furthermore,

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TABLE IV
COMPARISONS WITH ADVANCED OOD METHODS ON THE DIOR-R DATASET

Method	Backbone	Epoch	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	VM	mAP
One-stage:																							
FCOS-O [51]	R-50-FPN	12	48.70	24.88	63.57	80.97	18.41	68.99	23.26	42.37	60.25	64.83	60.66	31.84	40.80	73.09	66.32	56.61	77.55	38.10	30.69	55.87	51.39
RetinaNet-O [16]	R-50-FPN	12	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	65.57	61.09	81.46	47.33	38.01	60.24	57.55
Double-heads [52]	R-50-FPN	12	62.13	19.53	71.50	**87.09**	28.01	72.17	20.35	61.19	64.56	73.37	**81.97**	40.68	42.40	80.36	**73.12**	62.37	**87.09**	54.94	41.32	64.86	59.45
S^2A-Net [2]	R-50-FPN	12	65.40	42.04	75.15	83.91	36.01	72.61	28.01	65.09	75.11	75.56	80.47	35.91	52.10	82.33	65.89	66.08	84.61	54.13	48.00	69.67	62.90
Oriented RepPoints [43]	R-50-FPN	40	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	66.71
Ours:																							
S^2A-Net + FRIoU	R-50-FPN	12	64.93	44.27	73.97	83.89	36.92	72.56	29.16	64.96	73.66	75.64	79.91	38.75	53.68	83.18	65.69	**68.56**	85.91	56.27	49.34	66.31	63.38
S^2A-Net + FRIoU	R-50-FPN	36	**67.07**	**54.37**	**75.25**	85.64	**43.56**	**77.11**	**32.39**	**71.05**	**79.99**	**78.18**	80.42	**46.30**	**58.70**	**85.57**	71.35	68.23	85.13	**58.08**	**52.18**	**70.85**	**67.07**
Two-stage:																							
Faster RCNN-O [46]	R-50-FPN	12	62.79	26.80	71.72	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21	59.54
Gliding Vertex [49]	R-50-FPN	12	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
RoI Trans. [31]	R-50-FPN	12	63.34	37.88	71.78	87.53	40.68	72.60	26.86	78.71	68.09	68.96	**82.74**	47.71	55.61	81.21	**78.23**	70.26	81.61	54.86	43.27	65.52	63.87
AOPG [39]	R-50-FPN	12	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	52.38	69.99	64.41
Oriented R-CNN [30]	R-50-FPN	12	62.00	44.92	71.78	87.93	43.84	72.64	35.46	66.39	81.35	74.10	80.95	43.52	58.42	81.25	68.01	62.52	**88.62**	59.31	43.27	66.31	64.63
DODet [50]	R-50-FPN	12	63.40	43.35	72.11	81.32	43.12	72.59	33.32	**78.77**	70.84	74.15	75.47	48.00	59.31	85.41	74.04	71.56	81.52	55.47	51.86	66.40	65.10
Ours:																							
AOPG + FRIoU	R-50-FPN	12	**70.19**	44.91	**77.54**	87.78	40.95	72.61	34.12	65.53	80.30	76.57	81.62	47.61	56.07	89.16	72.91	**78.46**	88.50	59.79	53.79	69.99	67.42
AOPG + FRIoU	R-50-FPN	36	62.79	**54.62**	71.59	**88.41**	**47.26**	**77.68**	**40.09**	71.49	**84.20**	**81.46**	82.67	**53.45**	**60.47**	**89.62**	75.07	78.37	81.53	**62.80**	**56.28**	**71.97**	**69.59**

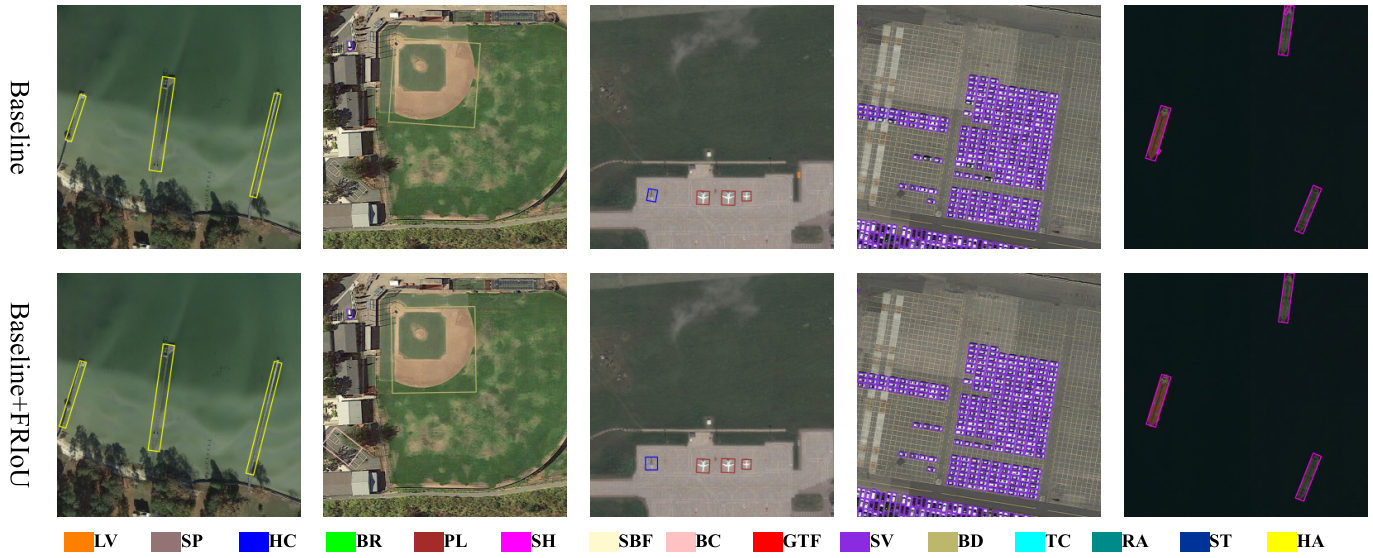


Fig. 6. Subjective comparisons between the baseline method [39] and the baseline + FRIoU on the DOTA dataset. One color stands for one object class. The same below.

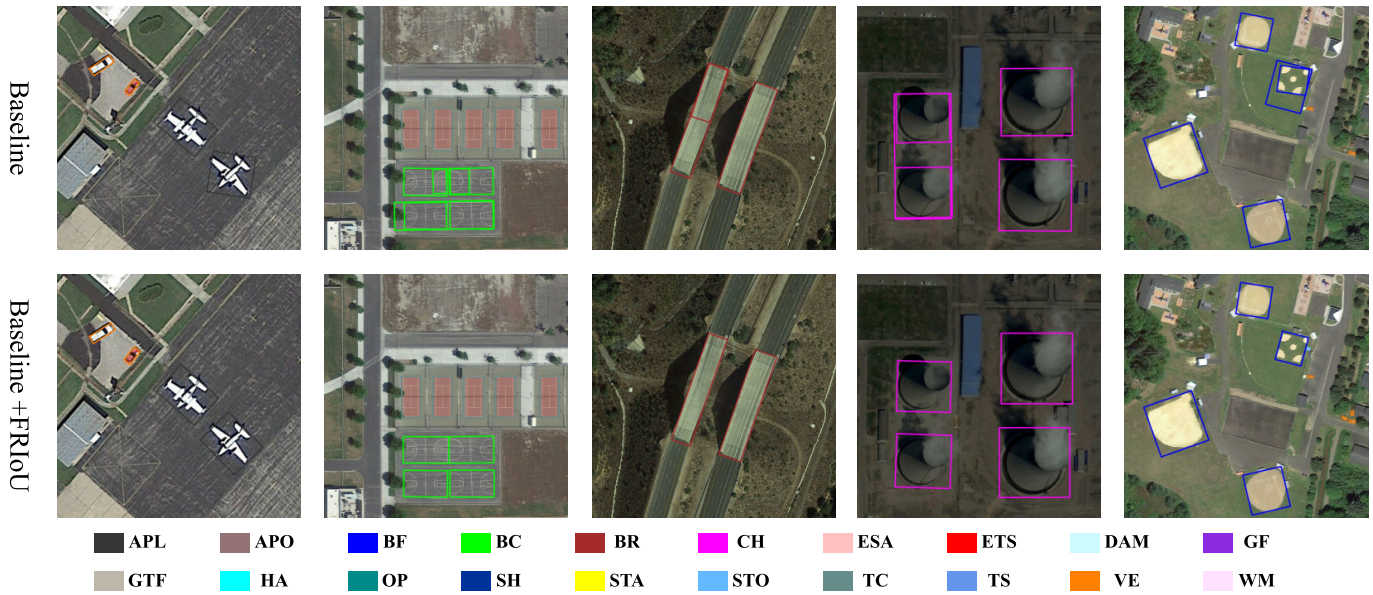


Fig. 7. Subjective comparisons between the baseline method and the baseline + FRIoU on the DIOR-R dataset.

S^2A-Net + FRIoU (AOPG + FRIoU) obtains the best results in 16 out of 20 categories.

In summary, our methods achieve SOTA performance on both the DOTA and DIOR-R datasets.

The subjective comparisons between the baseline method (AOPG) and baseline + FRIoU on the DOTA and DIOR-R datasets are shown in Figs. 6 and 7, respectively. Obviously, the detection results can be improved using FRIoU loss.

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V. Conclusion

A UTS was proposed to provide a universal solution which facilitates the transferring of BBR loss from HOD to OOD. Following the UTS, the RIoU loss was designed for BBR of OOD by transferring the EIoU loss in HOD. As a matter of fact, other BBR losses for HOD can also be transferred to OOD following the UTS, and the RIoU loss can be considered as an example to show how to transfer. On the basis of RIoU loss, a novel BBR loss named FRIoU loss was proposed to assign larger weights for hard samples in the processing of BBR. The comparisons with other BBR losses show that the proposed RIoU and FRIoU losses can give better performance, which validates the effectiveness of the proposed UTS. The ablation study of FRIoU loss shows that the performance of FRIoU loss is superior to RIoU loss, which indicates that giving more attention to hard samples in the processing of BBR can effectively increase the upper bound of the performance of OOD model. The validation experiment of generalization capability of FRIoU loss demonstrates that the performance of various types of OOD methods can be improved using FRIoU loss. The comprehensive quantitative comparisons with many advanced OOD methods show that the combination of baseline methods and FRIoU loss can achieve SOTA performance on two RSI datasets, which validates the effectiveness of FRIoU loss again.

Apparently, the performance of FRIoU loss is determined to a large extent by the capability of EIoU loss, and therefore, finding more powerful BBR loss in HOD is an important work in the future.

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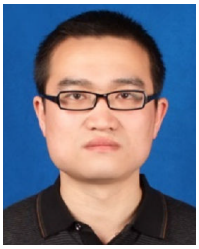
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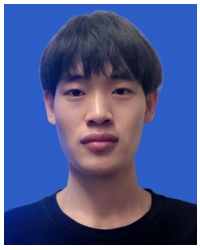
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